M22 Ketamine

Scope of Practice	EP
Classification	• Dissociative Anesthetic
Usual Presentation	• 500mg/10ml ampule
Mechanism of Action	• Ketamine predominantly blocks the action of glutamate on NMDA receptors, resulting in dose dependent analgesia, amnesia, disassociation and anesthesia.
Indications	• Sedation of agitated patients.
Contraindications	• Known allergy • Significant hypertension.
Precautions	• Active psychosis – ketamine may make this worse • Coadministration with haloperidol or benzodiazepines (enhanced sedative effect)
Side Effects	• Laryngospasm • Respiratory Depression (usually associated with large or rapid IV doses) • Emergence Phenomena • Increase airway secretions • Nystagmus • Transient hypertension
Adult Dose & Administration	**Sedation for Abnormal Behavior** • IM: 1-4mg/kg (repeat after 10 minutes, once only) • IV/IO: 1mg/kg bolus over 1-2minutes. Maximum dose 100mg. **Sedation for Agitation** (e.g. ROSC/TBI/Cardioversion/Pacing) • 0.25-1mg/kg IV/IO **Dilution for IV Administration** • In a 10ml Syringe, draw up 8mls of normal saline. • Add 2ml (100mg) of Ketamine. • Final concentration of the solution is 10mg/1ml
Pediatric Dose & Administration	• Not for pediatric administration
Use in Pregnancy and Lactation	• Pregnancy: Safety has not been demonstrated. Administration in pregnancy is not recommended. • Lactation: Safety has not been demonstrated. Administration in lactation is not recommended.

Additional Information	• Ketamine must not be administered to facilitate intubation.
	• Many of the adverse effects of ketamine are related to high doses or rapid IV administration.
	  • Where possible slowly administer IV doses slowly.
	• Emergence Phenomena only requires management if it is impeding patient care.
	  • Low dose benzodiazepines may be administered if required however this may cause significant lowering of consciousness.
	  • Benzodiazepines should be administered IV to enable lower and more accurate dosages.